# II-VOLUMETRIC ANALYSIS

## 1. Estimation of Ferrous Sulphate ($Fe^{2+}$)

### Aim :

To estimate the amount of ferrous sulphate dissolved in 750 ml of the given unknown solution volumetrically. For this you are given with a standard solution of ferrous ammonium sulphate (FAS) of normality 0.1102 N and potassium permanganate solution as link solution.

### Principle:

During these titrations, $Fe^{2+}$ ions (from ferrous salts) are oxidised to $MnO_4^-$ ions and $MnO_4^-$ ion (from $Mn^{2+}$) is reduced to $Mn^{2+}$ ion.

**Oxidation** : $5\ Fe^{2+} \longrightarrow 5\ Fe^{3+} + 5e^-$

**Reduction** : $\underset{\text{Pink}}{MnO_4^-} + 8H^+ + 5e^- \longrightarrow \underset{\text{colourless}}{Mn^{2+}} + 4H_2O$

**Overall reaction** : $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

**Short procedure:**

| S.no | Content | Titration-I | Titration-II |
|---|---|---|---|
| 1 | Burette solution | $KMnO_4$ | $KMnO_4$ |
| 2 | Pipette solution | 20 ml of standard FAS | 20 ml of unknown $FeSO_4$ |
| 3 | Acid to be added | 20ml of 2N $H_2SO_4$ (approx) | 20ml of 2N $H_2SO_4$ (approx) |
| 4 | Temperature | Lab temperature | Lab temperature |
| 5 | Indicator | Self-indicator ($KMnO_4$) | Self-indicator ($KMnO_4$) |
| 6 | End point | Appearance of permanent pale pink colour | Appearance of permanent pale pink colour |
| 7 | Equivalent weight of $FeSO_4$ = 278 | | |

### Procedure :

### Titration–I

(Link $KMnO_4$)Vs (Standard FAS)

Burette is washed with water, rinsed with $KMnO_4$ solution and filled with same $KMnO_4$ solution up to the zero mark. Exactly 20 ml of standard FAS solution is pipetted out into the clean, washed conical flask. To this FAS solution, approximately 20ml of 2N sulphuric acid is added. This mixture is titrated against $KMnO_4$ Link solution from the burette. $KMnO_4$ is added drop wise till the appearance of permanent pale pink colour. Burette reading is noted, and the same procedure is repeated to get concordant values.



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